

Mock Paper 1 — Computer Systems (H446/01)

Time allowed: 2 hours 30 minutes · Total: 140 marks · Answer all questions.

Instructions: Show all working on calculations and logic questions. Mark against the mark scheme at the end.

AO key: **AO1** = knowledge & understanding · **AO2** = application · **AO3** = analysis/evaluation/design.

Running mark tally is shown beside each question. Whole-paper total is verified at the very end.

Section A / questions

Question 1 — Processor architecture and the FDE cycle (8 marks)

A CPU uses the Von Neumann architecture and contains the registers PC, MAR, MDR, CIR and ACC.

1(a) State the role of the **Program Counter (PC)** and the **Memory Address Register (MAR)** during the fetch stage. [3]

1(b) Describe **three** factors, other than clock speed, that affect the performance of a processor. [3]

1(c) Explain **one** advantage of a Harvard architecture over a Von Neumann architecture for an embedded system. [2]

(Running total: 8)

Question 2 — Storage devices (7 marks)

2(a) State **two** characteristics of a solid-state drive (SSD) that make it suitable for use in a laptop. [2]

2(b) Compare optical storage with magnetic storage, giving **three** distinct points of comparison. [3]

2(c) A digital camera stores photographs on a removable memory card. State the **type** of storage technology used and justify why it is appropriate. [2]

(Running total: 15)

Question 3 — Operating systems (9 marks)

3(a) Explain how a **paging** memory-management technique allows a program larger than physical RAM to run. [4]

3(b) Describe the purpose of an **interrupt** and explain the role of the **interrupt service routine (ISR)**. [3]

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3(c) State **one** difference between a **multi-tasking** and a **multi-user** operating system. [2]

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(Running total: 24)

Question 4 — Translators and the compilation process (9 marks)

4(a) State **three** differences between a **compiler** and an **interpreter**. [3]

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4(b) The compilation of source code passes through four stages: lexical analysis, syntax analysis, code generation and optimisation. Describe what happens during **lexical analysis** and **syntax analysis**. [4]

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4(c) Explain why a programmer might choose to distribute software as **bytecode** for a virtual machine rather than as native machine code. [2]

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(Running total: 33)

Question 5 — Software development methodologies (7 marks)

5(a) A start-up is building a mobile app where the client's requirements are expected to change frequently. Justify why an **Agile** methodology would be more suitable than the **Waterfall** model. [4]

5(b) Describe **three** features of the **Extreme Programming (XP)** methodology. [3]

(Running total: 40)

Question 6 — Object-oriented programming (9 marks)

A program models library members. A base class `Member` has the attributes `name` and `memberID`. A subclass `StudentMember` adds the attribute `course` and overrides the method `borrowLimit()`.

6(a) Explain what is meant by **inheritance** and **encapsulation**, using this scenario to illustrate each. [3]

6(b) Write the class definition for `Member` in pseudocode (or a language of your choice), including a constructor and a `getDetails()` method. Use private attributes with public accessor methods. [4]

6(c) Explain what is meant by **polymorphism** and give an example using `borrowLimit()`. [2]

(Running total: 49)

Question 7 — Assembly language and addressing modes (8 marks)

A processor uses the Little Man Computer (LMC)-style instruction set. Consider this program:

```
      INP
      STA num
loop  LDA total
      ADD num
      STA total
      LDA num
      SUB one
      STA num
      BRP loop
      LDA total
      OUT
      HLT
num   DAT
total DAT 0
one   DAT 1
```

7(a) Trace the program for an input of **3**. State the final value output and explain in one sentence what the program computes. [4]

7(b) Explain the difference between **immediate** and **direct (absolute)** addressing, and state **one** reason a processor also provides **indexed** addressing. [4]

(Running total: 57)

Question 8 — Compression, encryption and hashing (10 marks)

8(a) Explain the difference between **lossy** and **lossless** compression, giving **one** suitable use case for each. [3]

8(b) Describe how **run-length encoding (RLE)** compresses data and state **one** type of data for which it is ineffective. [3]

8(c) Explain how **symmetric** encryption differs from **asymmetric** encryption, and describe how asymmetric encryption can be used to create a **digital signature**. [4]

(Running total: 67)

Question 9 — Databases and SQL (10 marks)

A **Booking** table and a **Customer** table store data for a car-hire company:

```
Customer(CustomerID, Surname, City)
Booking(BookingID, CustomerID, CarReg, StartDate, Days)
```

9(a) State what is meant by a **primary key** and a **foreign key**, identifying one of each from the tables above. [2]

9(b) Write an SQL statement to list the **Surname** and **City** of every customer who lives in **'Leeds'**, sorted alphabetically by **Surname**. [3]

9(c) Write an SQL statement that lists the **Surname** of each customer together with the **CarReg** and **StartDate** of all their bookings of **5 or more** days. [5]

(Running total: 77)

Question 10 — Networks and TCP/IP (10 marks)

10(a) Describe the role of each of the four layers of the **TCP/IP stack**. [3]

10(b) Explain how the **Domain Name System (DNS)** resolves a URL into an IP address, referring to the hierarchy of DNS servers. [4]

10(c) Explain the difference between **circuit switching** and **packet switching**, and state why packet switching is used on the internet. [3]

(Running total: 87)

Question 11 — Number representation (12 marks)

11(a) Convert the denary number **-42** into **8-bit two's complement**. Show your working. [3]

11(b) A floating-point number uses an **8-bit mantissa** and a **4-bit exponent**, both in two's complement, with an assumed binary point after the first mantissa bit. The stored value is:

mantissa = 0101 1000 exponent = 0011

Calculate the denary value this represents. Show your working, and state whether the number is **normalised**, justifying your answer. [5]

11(c) Perform the binary addition 0110 1101 + 0101 0011 in 8 bits. State the 8-bit result and explain whether **overflow** has occurred if the values are interpreted as signed two's complement. [4]

(Running total: 99)

Question 12 — Data structures (10 marks)

12(a) Explain the difference between a **stack** and a **queue**, and give **one** real use of each in a computer system. [3]

12(b) A **circular queue** is implemented in an array of size 5 with **front** and **rear** pointers. Explain **why** a circular queue is preferred to a linear queue, and describe how the **rear** pointer wraps around. **[4]**

12(c) State the **Big O** time complexity of: (i) adding an item to the top of a stack; (ii) a linear search of an unsorted list of n items; (iii) a binary search of a sorted list of n items. **[3]**

(Running total: 109)

Question 13 — Boolean algebra and logic circuits (12 marks)

13(a) Complete the truth table for the expression $Q = (A \text{ AND NOT } B) \text{ OR } (\text{NOT } A \text{ AND } B)$, and state the name of the single logic gate that this expression is equivalent to. **[4]**

13(b) Using the laws of Boolean algebra, simplify the expression $Q = A \wedge B \vee A \wedge (B \vee C) \vee B \wedge (B \vee C)$ to its simplest form. Show each step and name the law used. **[4]**

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13(c) A safety system sounds an alarm ($Z = 1$) when **either** a door sensor **D** is open **and** the system is armed **A**, **or** when a panic button **P** is pressed (regardless of the other inputs). Write the Boolean expression for **Z** and draw the logic circuit using AND, OR and any other required gates. **[4]**

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(Running total: 121)

Question 14 — Networking and security (7 marks)

14(a) Explain the difference between a **LAN** and a **WAN**, referring to ownership and geographical scale. **[2]**

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14(b) Describe how a **firewall** using **packet filtering** helps protect a network. **[3]**

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14(c) Explain why a **star** network topology is more robust than a **bus** topology when a single connection fails. **[2]**

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(Running total: 128)

Question 15 — Extended response: legal, moral and ethical issues (12 marks)

A large retailer introduces an **AI-driven facial-recognition system** in its stores. The system scans the faces of every customer who enters, matches them against a database of known shoplifters, and alerts security staff in real time. Camera footage and biometric data are stored on the company's servers for 12 months.

15 Discuss the **legal, ethical and privacy issues** raised by this system. In your answer you should consider the interests of customers, the retailer and society, and reach a justified conclusion about whether the system should be deployed. **[12]**

(Running total: 140)

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